

Venkateswara Sahu

Applied AI and Machine Learning Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Open to relocation

EDUCATION

Lovely Professional University

Aug 2022 – May 2026

B.Tech (Hons.) Computer Science and Engineering - Data Science and Data Engineering

Punjab, India

CGPA: 8.38/10

TECHNICAL SKILLS

Languages: Python, SQL, Java, C/C++

Machine Learning: PyTorch, scikit-learn, XGBoost, LightGBM, Optuna

Generative AI: LangGraph, LangChain, RAG, FAISS, sentence-transformers, LLM APIs

Engineering: FastAPI, Flask, Docker, GitHub Actions, Kafka, Airflow, MLflow, TiDB Cloud

EXPERIENCE

TransOrg Analytics (Pickl.AI) - LPU Industry Tie-Up

Punjab, India

Generative AI Intern

Jan 2026 – May 2026

- Led the technical implementation of F1InsightAI, a nine-node LangGraph pipeline that translated natural-language questions into SQL across 700,000+ Formula 1 records in 14 TiDB tables.
- Achieved 83.3% first-attempt SQL accuracy (15/18 queries) and improved average schema-retrieval Mean Reciprocal Rank from 0.12 to 0.67 across three evaluation iterations.
- Implemented SQL validation and error-guided retry paths, then exposed the workflow through a Dockerized Flask API and a public [Hugging Face demo](#) with evaluation telemetry.

PROJECTS

Vigil (vigil-drift) - Unsupervised Concept Drift Detection

Jun–Aug 2026 | [GitHub](#) | [PyPI](#)

Python, PyTorch, Kafka, Airflow, MLflow, FastAPI

- Published a reusable Python package that detects distribution shifts without labels using adaptive and frozen autoencoders, reconstruction-error tests, and kernel density estimation.
- Built reproducible attribution comparisons across synthetic shifts and controlled CICIDS2017 development data; documented failure modes against statistical baselines.
- Added automated regression tests, package installation checks and GitHub Actions, plus Kafka ingestion and Airflow retraining examples.

P&ID Document Intelligence and Material Take-Off Extraction

Jul 2026 | [GitHub](#)

Python, YOLOv8, OCR, NetworkX, LangGraph

- Combined symbol detection, connected-component-guided OCR, and spatial graph matching to convert engineering drawings into structured equipment and instrumentation records.
- Reduced OCR processing time from approximately 360 seconds to 7 seconds in project tests, and added rule-based validation that flags low-confidence entity relationships for review.

CTR Prediction and Ad Scoring System

Sep–Nov 2025 | [GitHub](#) | [Live Demo](#)

Python, XGBoost, LightGBM, Optuna, Flask, Streamlit

- Engineered 150 features from 39 raw fields across 10 million Criteo advertising records and used Optuna to tune XGBoost and LightGBM models.
- Achieved 0.9067 test AUC with XGBoost versus a 0.9024 LightGBM baseline, then served predictions through a Flask API and interactive dashboard with sub-500 ms inference in project tests.

LEADERSHIP

Unitech Startup

Punjab, India

Seed Money Project Lead

Mar 2024 – May 2026

- Secured INR 100,000 in university seed funding by presenting the product concept, implementation roadmap, and budget to an academic evaluation panel.